

DEXTER FERNANDES

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SUMMARY

Robotics and perception engineer with an MSc in Robotics (Merit) and over three years of commercial experience in real-time C++ and Python systems. Built LiDAR SLAM and multi-sensor localisation in C++17 and ROS2 on a Clearpath Husky, fusing LiDAR, depth and IMU sensing, and benchmarked four SLAM stacks under one evaluation protocol. Shipped real-time perception to 24/7 outdoor industrial sites on constrained edge hardware, with on-site field validation, drift tracking and measured reliability gains. Comfortable across Nav2 navigation stacks, sensor calibration, point cloud processing and deterministic real-time inference.

RELEVANT EXPERIENCE

Esbaar – AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Developed and maintained NVIDIA DeepStream 6.x and FFmpeg video pipelines for 24/7 perception on BP drilling rigs, covering camera ingestion, inference and field diagnostics in desert conditions.
- Owned the end-to-end ML lifecycle for real-time perception on BP rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Hardened deployed services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Built a watchdog application with a REST API exposing service health and restart control, so failed edge services recovered remotely without an engineer on site.
- Led on-site validation on BP rig deployments, diagnosing model failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.
- Owned post-deployment monitoring and drift tracking for fielded vision systems, catching regressions early and feeding failure analysis back into retraining.
- Added observability with structured logs, metrics, timeouts, retries and message validation, cutting root-cause analysis time on live deployments.
- Used multiprocessing to isolate capture from inference on shared edge hardware, keeping one stalled stream from blocking the rest of the pipeline.

LivNSense Technologies – Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, raising throughput from 5 to 15 FPS under real-time constraints.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and running incident detection under edge compute constraints.
- Implemented optical flow based ego-motion estimation feeding CNN classifiers in PyTorch to infer vehicle state, supporting collision avoidance on moving vehicles.
- Added ByteTrack multi-object tracking over detections to hold worker and vehicle identities across frames, cutting duplicate incident alerts from the moving-vehicle feeds.
- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Profiled C++ inference and pre/post-processing hotspots with Intel VTune, overlapping preprocessing, inference and post-processing to cut end-to-end latency.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Wrote gtest unit tests and documented interfaces for the C++ inference path, catching regressions before deployment.

PROJECTS

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection.
- Ran the Nav2 stack with NavFn Dijkstra global planning into an MPPI sampling-based model predictive controller, evaluating path smoothing and recovery-behaviour handling on waypoint routes.
- Developed ROS2 nodes, launch files and parameterised configs in C++ and Python, using tf2 and message synchronisation to debug transform timing, and added an EKF localisation layer with robot_localization fusing wheel odometry and IMU.

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.

EDUCATION

University of Bristol – MSc Robotics, Merit

09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (roslcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing, running the full stack on an NVIDIA Jetson AGX Xavier within the platform's compute and memory limits.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline, configuring Ceres, g2o and GTSAM as solver backends and tuning ICP parameters across environments.
- Configured Nav2 in simulation for autonomous waypoint traversal over the generated map, running NavFn Dijkstra global planning into a DWB (Dynamic Window Approach) local controller, testing survey routes before running them on the Husky.

Visvesvaraya National Institute of Technology – BTech Electrical and Electronics Engineering 07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	C++17, Python, CUDA C++, Bash, SQL
Robotics and ROS2	ROS2 (Robot Operating System 2), roslcpp, roslpy, tf2, rosbag2, launch files, parameterised configs, Gazebo, Gazebo Harmonic, message synchronisation, transform timing debugging
Localisation and SLAM	LiDAR SLAM, sensor fusion, RTAB-Map, LIO-SAM, Cartographer, GLIM, ICP (Iterative Closest Point), scan matching, pose graph optimisation, EKF, robot_localization, wheel odometry and IMU fusion, LiDAR extrinsics calibration, PCL (Point Cloud Library), Open3D, Eigen, Ceres, g2o, GTSAM
Navigation and control	Nav2, global and local path planning, NavFn, Dijkstra, DWB (Dynamic Window Approach), MPPI (Model Predictive Path Integral, sampling-based MPC), path smoothing, behaviour trees, recovery behaviours, waypoint following
Sensors and platforms	Velodyne VLP-16 LiDAR, Intel RealSense D435i, IMU, multi-camera synchronisation, RGB and IR camera and ISP tuning, Clearpath Husky, NVIDIA Jetson AGX Xavier, JetPack, Intel Movidius VPU
Real-time and optimisation	TensorRT (custom plugins, dynamic batching), OpenVINO, ONNX, FP16 and INT8 quantisation, structured pruning, layer fusion, latency and throughput profiling, Intel VTune, multiprocessing, algorithm optimisation (optimization), deterministic real-time inference
Perception and ML	object detection, object tracking, optical flow, ego-motion estimation, semantic segmentation, convolutional neural networks (CNNs), YOLO, ByteTrack, PyTorch, OpenCV, NumPy, TIDE error analysis, failure-mode analysis, drift tracking
Engineering practice	Git, CMake, Docker, CI/CD, pytest, gtest, Linux, benchmarking harnesses, ablation studies, Agile and Scrum, technical leadership, mentoring, stakeholder communication

ADDITIONAL EXPERIENCE

GrowthStage – Technical Assessor, Computer Vision and Software Engineering 05/2026 – PRESENT · REMOTE
Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

Taco Bell – Service Champion / Team Trainer 09/2024 – PRESENT · BRISTOL, ENGLAND
Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)